

DBMS QUESTIONS

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Course : B.Tech CSE AIML

Section : B

Sem : III

Q1. Display each employee's salary along with the total salary of their department.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- college
- college_management
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Emp_Name, Dept_ID, Salary,
2    SUM(Salary) OVER(PARTITION BY Dept_ID) AS Dept_Total
3 FROM Employee;
```

Result Grid

	Emp_Name	Dept_ID	Salary	Dept_Total
▶	Ravi	D1	90000	225000
	Priya	D1	75000	225000
	Sneha	D1	60000	225000
	Karan	D2	55000	100000
	Amit	D2	45000	100000
	Neha	D3	82000	130000
	Arjun	D3	48000	130000
	Divya	D4	48000	178000
	Vikas	D4	65000	178000
	Meera	D4	65000	178000

Administration Schemas

Information

No object selected

Result 2

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	16:56:58	CREATE TABLE Department (Dept_ID VARCHAR(10) PRIMARY KE...	0 row(s) affected	0.078 sec
4	16:56:58	CREATE TABLE Employee (Emp_ID INT PRIMARY KEY, Emp_Na...	0 row(s) affected	0.015 sec
5	16:57:10	INSERT INTO Department VALUES ('D1', 'IT'), ('D2', 'HR'), ('D3', 'Financ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
6	16:57:21	INSERT INTO Employee VALUES (101, 'Ravi', 'D1', 90000, '2016-03-1...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
7	16:57:29	SELECT * FROM Employee LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
8	17:01:27	SELECT Emp_Name, Dept_ID, Salary, SUM(Salary) OVER(PARTITI...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q2. Display each employee's salary and the average salary of their department.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1 x

```
1 SELECT Emp_Name, Dept_ID, Salary,
2     AVG(Salary) OVER(PARTITION BY Dept_ID) AS Dept_Avg
3 FROM Employee;
```

Result Grid

Emp_Name	Dept_ID	Salary	Dept_Avg
Ravi	D1	90000	75000.0000
Priya	D1	75000	75000.0000
Sneha	D1	60000	75000.0000
Karan	D2	55000	50000.0000
Amit	D2	45000	50000.0000
Neha	D3	82000	65000.0000
Arjun	D3	48000	65000.0000
Divya	D4	48000	59333.3333
Vikas	D4	65000	59333.3333
Meera	D4	65000	59333.3333

Administration Schemas

Information

Schema: tanwar_collections

Result 5 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	17:03:28	use tanwar_collections	0 row(s) affected	0.000 sec
14	17:03:28	Select * from products LIMIT 0, 1000	6 row(s) returned	0.016 sec / 0.000 sec
15	17:03:38	use tanwar_collections	0 row(s) affected	0.000 sec
16	17:03:38	Select * from users LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
17	17:04:00	use companydb	0 row(s) affected	0.000 sec
18	17:05:20	SELECT Emp_Name, Dept_ID, Salary, AVG(Salary) OVER(PARTITI...	10 row(s) returned	0.000 sec / 0.000 sec

Q3. Find what percentage of the total department salary each employee earns.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 • SELECT Emp_Name, Dept_ID, Salary,
2     ROUND(Salary * 100.0 /
3     SUM(Salary) OVER(PARTITION BY Dept_ID), 2) AS Pct_Of_Dept
4 FROM Employee;
```

Result Grid

Emp_Name	Dept_ID	Salary	Pct_Of_Dept
Ravi	D1	90000	40.00
Priya	D1	75000	33.33
Sneha	D1	60000	26.67
Karan	D2	55000	55.00
Amit	D2	45000	45.00
Neha	D3	82000	63.08
Arjun	D3	48000	36.92
Divya	D4	48000	26.97
Vikas	D4	65000	36.52
Meera	D4	65000	36.52

Administration Schemas

Information

No object selected

Result 6

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	17:03:28	Select * from products LIMIT 0, 1000	6 row(s) returned	0.016 sec / 0.000 sec
✓ 15	17:03:38	use tanwar_collections	0 row(s) affected	0.000 sec
✓ 16	17:03:38	Select * from users LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 17	17:04:00	use companydb	0 row(s) affected	0.000 sec
✓ 18	17:05:20	SELECT Emp_Name, Dept_ID, Salary, AVG(Salary) OVER(PARTITI...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 19	17:06:32	SELECT Emp_Name, Dept_ID, Salary, ROUND(Salary * 100.0 / ...	10 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q4. Give every employee a unique row number according to salary from highest to lowest.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Emp_Name, Salary,
2     Execute the selected portion of the script or everything, if there is no selection
3 FROM Employee;
```

Result Grid

Emp_Name	Salary	Row_Num
Ravi	90000	1
Neha	82000	2
Priya	75000	3
Vikas	65000	4
Meera	65000	5
Sneha	60000	6
Karan	55000	7
Arjun	48000	8
Divya	48000	9
Amit	45000	10

Administration Schemas

Information

No object selected

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Form Editor

Field Types

Query Stats

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	17:03:38	use tanwar_collections	0 row(s) affected	0.000 sec
16	17:03:38	Select * from users LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
17	17:04:00	use companydb	0 row(s) affected	0.000 sec
18	17:05:20	SELECT Emp_Name, Dept_ID, Salary, AVG(Salary) OVER(PARTITI...	10 row(s) returned	0.000 sec / 0.000 sec
19	17:06:32	SELECT Emp_Name, Dept_ID, Salary, ROUND(Salary * 100.0 / ...	10 row(s) returned	0.016 sec / 0.000 sec
20	17:07:08	SELECT Emp_Name, Salary, ROW_NUMBER() OVER(ORDER BY ...	10 row(s) returned	0.000 sec / 0.000 sec

Q5. Rank all employees according to salary from highest to lowest.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Emp_Name, Salary,
2 RANK() OVER(ORDER BY Salary DESC) AS Salary_Rank
3 FROM Employee;
```

Result Grid

Emp_Name	Salary	Salary_Rank
Ravi	90000	1
Neha	82000	2
Priya	75000	3
Vikas	65000	4
Meera	65000	4
Sneha	60000	6
Karan	55000	7
Arjun	48000	8
Divya	48000	8
Amit	45000	10

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	17:05:20	SELECT Emp_Name, Dept_ID, Salary, AVG(Salary) OVER(PARTITI...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 19	17:06:32	SELECT Emp_Name, Dept_ID, Salary, ROUND(Salary * 100.0 / ...	10 row(s) returned	0.016 sec / 0.000 sec
✓ 20	17:07:08	SELECT Emp_Name, Salary, ROW_NUMBER() OVER(ORDER BY ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 21	17:07:38	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DES...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 22	17:07:54	SELECT Emp_Name, Salary, ROW_NUMBER() OVER(ORDER BY ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 23	17:07:58	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DES...	10 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q6. Rank employees according to salary using dense ranking.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Emp_Name, Salary,  
2 DENSE_RANK() OVER(ORDER BY Salary DESC) AS dr  
3 FROM Employee;
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

	Emp_Name	Salary	dr
▶	Ravi	90000	1
	Neha	82000	2
	Priya	75000	3
	Vikas	65000	4
	Meera	65000	4
	Sneha	60000	5
	Karan	55000	6
	Arjun	48000	7
	Divya	48000	7
	Amit	45000	8

Administration Schemas

Information

No object selected

Result 11

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 21	17:07:38	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DESC)	10 row(s) returned	0.000 sec / 0.000 sec
✓ 22	17:07:54	SELECT Emp_Name, Salary, ROW_NUMBER() OVER(ORDER BY ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 23	17:07:58	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DESC)	10 row(s) returned	0.000 sec / 0.000 sec
✗ 24	17:08:32	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
✗ 25	17:09:16	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
✓ 26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Q7. Divide employees into 4 groups according to salary

MySQL Workbench

Local instance MySQL80

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Navigator: SCHEMAS

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 • SELECT Emp_Name, Salary,  
2     NTILE(4) OVER(ORDER BY Salary DESC) AS Quartile  
3 FROM Employee;
```

Result Grid

	Emp_Name	Salary	Quartile
▶	Ravi	90000	1
	Neha	82000	1
	Priya	75000	1
	Vikas	65000	2
	Meera	65000	2
	Sneha	60000	2
	Karan	55000	3
	Arjun	48000	3
	Divya	48000	4
	Amit	45000	4

Information: No object selected

Output: Action Output

#	Time	Action	Message
✓ 22	17:07:54	SELECT Emp_Name, Salary, ROW_NUMBER() OVER(ORDER BY ...	10 row(s) returned
✓ 23	17:07:58	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DES...	10 row(s) returned
✗ 24	17:08:32	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check t...
✗ 25	17:09:16	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check t...
✓ 26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned
✓ 27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Snipping Tool

Screenshot copied to clipboard. Automatically saved to screen.

Q8. Rank employees separately inside each department based on salary.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Dept_ID, Emp_Name, Salary,  
2     RANK() OVER(  
3         PARTITION BY Dept_ID  
4         ORDER BY Salary DESC  
5     ) AS Dept_Rank  
6 FROM Employee;
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Dept_ID	Emp_Name	Salary	Dept_Rank
D1	Ravi	90000	1
D1	Priya	75000	2
D1	Sneha	60000	3
D2	Karan	55000	1
D2	Amit	45000	2
D3	Neha	82000	1
D3	Arjun	48000	2
D4	Vikas	65000	1
D4	Meera	65000	1
D4	Divya	48000	3

Administration Schemas

Information

No object selected

Result 13

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	17:07:58	SELECT Emp_Name, Salary, RANK() OVER(ORDER BY Salary DES...	10 row(s) returned	0.000 sec / 0.000 sec
24	17:08:32	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
25	17:09:16	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned	0.000 sec / 0.000 sec
27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned	0.015 sec / 0.000 sec
28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Q9. Show each employee's salary and the salary of the employee hired immediately before them.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 • SELECT Emp_Name, Hire_Date, Salary,
2     LAG(Salary) OVER(ORDER BY Hire_Date) AS Prev_Salary
3 FROM Employee
4 ORDER BY Hire_Date;
```

Result Grid

Emp_Name	Hire_Date	Salary	Prev_Salary
Ravi	2016-03-15	90000	NULL
Karan	2016-05-05	55000	90000
Neha	2017-11-20	82000	55000
Priya	2018-07-01	75000	82000
Vikas	2019-04-18	65000	75000
Amit	2020-01-10	45000	65000
Divya	2020-09-12	48000	45000
Sneha	2021-02-14	60000	48000
Meera	2021-06-25	65000	60000
Arjun	2022-08-30	48000	65000

Result 14

Output

Action Output

#	Time	Action	Message	Duration / Fetch
24	17:08:32	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
25	17:09:16	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned	0.000 sec / 0.000 sec
27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned	0.015 sec / 0.000 sec
28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec
29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q10. Show each employee's salary and the salary of the employee hired immediately after them.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 • SELECT Emp_Name, Hire_Date, Salary,
2     LEAD(Salary) OVER(ORDER BY Hire_Date) AS Next_Salary
3 FROM Employee
4 ORDER BY Hire_Date;
```

Result Grid

	Emp_Name	Hire_Date	Salary	Next_Salary
▶	Ravi	2016-03-15	90000	55000
	Karan	2016-05-05	55000	82000
	Neha	2017-11-20	82000	75000
	Priya	2018-07-01	75000	65000
	Vikas	2019-04-18	65000	45000
	Amit	2020-01-10	45000	48000
	Divya	2020-09-12	48000	60000
	Sneha	2021-02-14	60000	65000
	Meera	2021-06-25	65000	48000
	Arjun	2022-08-30	48000	NULL

Administration Schemas

Information

No object selected

Result 15

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	17:09:16	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	Error Code: 1064. You have an error in your SQL syntax; check the manu...	0.000 sec
26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned	0.000 sec / 0.000 sec
27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned	0.015 sec / 0.000 sec
28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec
29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec
30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q11. Find the difference between each employee's salary and the salary of the previously hired employee.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Emp_Name, Hire_Date, Salary,
2     LAG(Salary) OVER(ORDER BY Hire_Date) AS Prev_Salary,
3     Salary - LAG(Salary) OVER(ORDER BY Hire_Date) AS Salary_Difference
4 FROM Employee
5 ORDER BY Hire_Date;
```

Result Grid

Emp_Name	Hire_Date	Salary	Prev_Salary	Salary_Difference
Ravi	2016-03-15	90000	NULL	NULL
Karan	2016-05-05	55000	90000	-35000
Neha	2017-11-20	82000	55000	27000
Priya	2018-07-01	75000	82000	-7000
Vikas	2019-04-18	65000	75000	-10000
Amit	2020-01-10	45000	65000	-20000
Divya	2020-09-12	48000	45000	3000
Sneha	2021-02-14	60000	48000	12000
Meera	2021-06-25	65000	60000	5000
Arjun	2022-08-30	48000	65000	-17000

Result 16

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 26	17:10:22	SELECT Emp_Name, Salary, DENSE_RANK() OVER(ORDER BY S...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned	0.015 sec / 0.000 sec
✓ 28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec
✓ 30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 31	17:13:08	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Object Info Session

Q12. Display the highest-paid employee of each department on every row.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 • SELECT Dept_ID, Emp_Name, Salary,  
2     FIRST_VALUE(Emp_Name) OVER(  
3         PARTITION BY Dept_ID  
4         ORDER BY Salary DESC  
5     ) AS Top_Earner  
6 FROM Employees;
```

Result Grid

Dept_ID	Emp_Name	Salary	Top_Earner
D1	Ravi	90000	Ravi
D1	Priya	75000	Ravi
D1	Sneha	60000	Ravi
D2	Karan	55000	Karan
D2	Amit	45000	Karan
D3	Neha	82000	Neha
D3	Arjun	48000	Neha
D4	Vikas	65000	Vikas
D4	Meera	65000	Vikas
D4	Divya	48000	Vikas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 27	17:10:50	SELECT Emp_Name, Salary, NTILE(4) OVER(ORDER BY Salary DE...	10 row(s) returned	0.015 sec / 0.000 sec
✓ 28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec
✓ 30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 31	17:13:08	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 32	17:13:38	SELECT Dept_ID, Emp_Name, Salary, FIRST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q13. Display the lowest-paid employee in each department on every row.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 SELECT Dept_ID, Emp_Name, Salary,  
2     LAST_VALUE(Emp_Name) OVER  
3     PARTITION BY Dept_ID  
4     ORDER BY Salary ASC  
5     ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING  
6     ) AS Lowest_Earner  
7 FROM Employee;
```

Result Grid

Dept_ID	Emp_Name	Salary	Lowest_Earner
D1	Sneha	60000	Ravi
D1	Priya	75000	Ravi
D1	Ravi	90000	Ravi
D2	Amit	45000	Karan
D2	Karan	55000	Karan
D3	Arjun	48000	Neha
D3	Neha	82000	Neha
D4	Divya	48000	Meera
D4	Vikas	65000	Meera
D4	Meera	65000	Meera

Result 18

Output

Action Output

#	Time	Action	Message	Duration / Fetch
28	17:11:34	SELECT Dept_ID, Emp_Name, Salary, RANK() OVER(PART...	10 row(s) returned	0.000 sec / 0.000 sec
29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec
30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec
31	17:13:08	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.000 sec / 0.000 sec
32	17:13:38	SELECT Dept_ID, Emp_Name, Salary, FIRST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec
33	17:14:07	SELECT Dept_ID, Emp_Name, Salary, LAST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q14. Display the employee hierarchy starting from employees who have no manager.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Query 1

```
1 WITH RECURSIVE EmployeeTree AS
2
3     SELECT Emp_ID, Emp_Name, Manager_ID, 1 AS Level
4     FROM Employee
5     WHERE Manager_ID IS NULL
6     UNION ALL
7     SELECT e.Emp_ID, e.Emp_Name, e.Manager_ID, t.Level + 1
8     FROM Employee e
9     JOIN EmployeeTree t
10    ON e.Manager_ID = t.Emp_ID
11 )
12 SELECT *
13 FROM EmployeeTree
14 ORDER BY Level, Emp_ID;
```

Result Grid

	Emp_ID	Emp_Name	Manager_ID	Level
▶	101	Ravi	NULL	1
	104	Karan	NULL	1
	106	Neha	NULL	1
	108	Divya	NULL	1
	102	Priya	101	2
	105	Amit	104	2
	107	Arjun	106	2
	109	Vikas	108	2

Result 19

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 29	17:12:06	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.015 sec / 0.000 sec
✓ 30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 31	17:13:08	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 32	17:13:38	SELECT Dept_ID, Emp_Name, Salary, FIRST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 33	17:14:07	SELECT Dept_ID, Emp_Name, Salary, LAST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 34	17:14:49	WITH RECURSIVE EmployeeTree AS (SELECT Emp_ID, Emp_Name...	10 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Object Info Session

Q15. Find the lowest-paid employee in every department and increase their salary by 10%.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
- companydb
- sakila
- sys
- tanwar_collections
- world

Administration Schemas

Information

No object selected

Query 1

```
1 WITH Lowest AS
2 (
3     SELECT Emp_ID,
4            RANK() OVER(
5                PARTITION BY Dept_ID
6                ORDER BY Salary ASC) AS rnk
7     FROM Employee
8 )
9 UPDATE Employee e
10 JOIN Lowest l
11 ON e.Emp_ID = l.Emp_ID
12 SET e.Salary = e.Salary * 1.10
13 WHERE l.rnk = 1;
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 30	17:12:37	SELECT Emp_Name, Hire_Date, Salary, LEAD(Salary) OVER(ORDE...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 31	17:13:08	SELECT Emp_Name, Hire_Date, Salary, LAG(Salary) OVER(ORDER...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 32	17:13:38	SELECT Dept_ID, Emp_Name, Salary, FIRST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 33	17:14:07	SELECT Dept_ID, Emp_Name, Salary, LAST_VALUE(Emp_Name) ...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 34	17:14:49	WITH RECURSIVE EmployeeTree AS (SELECT Emp_ID, Emp_Name...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 35	17:16:56	WITH Lowest AS (SELECT Emp_ID, RANK() OVER(...	4 row(s) affected Rows matched: 4 Changed: 4 Warnings: 0	0.015 sec

Object Info Session